MaCVi@WACV 2027 Maritime Visual Odometry Challenge Factsheet

[Author 1], [Author 2], [Author 3]
[Affiliation, City, Country]

Instructions: how to submit the factsheet

This factsheet describes the contribution of each team in the Maritime Visual Odometry (VO) challenge at MaCVi@WACV 2027. Please address all aspects listed below as completely as you can. The factsheet, the code and the final entry on the leaderboard are used to verify the reported results and to decide on the final ranking.

1. **Who submits.** The top-3 teams on the final Maritime VO leaderboard are required to submit a factsheet. Other teams may submit and are listed if their entry checks out.
2. **Reproducibility is a must.** Results that cannot be reproduced are not ranked. Running your code on the released test frames must reproduce the JSON bundle of your final entry, and its score must match the leaderboard score within normal run-to-run variation.
3. **Detailed descriptions are much appreciated:** all models and experiments you tested, ablations, visualisations, things that did not work, external data, pretrained models, ensembles, and so on. Disclose all external data and pretrained models.
4. **Fill in one factsheet per team**, for the single entry you selected as your final entry in the dashboard.

What to send. E-mail the following to the VO organizers listed below, with all your team members in cc:

- **A compiled PDF** of the factsheet.
- **A** `.zip` with the LaTeX source files of the factsheet (`main.tex`, `ref.bib`, `images/`, including all figures).

Code and weights. Provide a public code repository, or a private one shared with the organizers. Its README must reproduce the JSON bundle of your final entry from the released test frames, with the model weights included in the repository or linked from it. Give the repository and weights link in Section 1 and the exact commands in Section 4.

Organizer	Email
Harin Nallaguntla (VO track lead)	hnallaguntla@ucsd.edu
Benjamin Kiefer	benjamin@lookout.team

E-mail guide.

To: hnallaguntla@ucsd.edu

Cc: benjamin@lookout.team; all your team members

Subject: MaCVi@WACV 2027 Maritime Visual Odometry Challenge - TEAM_NAME

The body of the e-mail should contain:

a) the challenge name (MaCVi@WACV 2027 Maritime Visual Odometry Challenge),

b) your team name (exactly as on macvi.org),

c) the team leader's name and e-mail address,

d) the rest of the team members,

e) the macvi.org user IDs of all team members, including the team leader,

f) the submission ID of your final entry and the user ID that uploaded it,

g) the author list in your team's author order: full names, affiliations, e-mails and OpenReview profile IDs,

h) the link to your code repository and to your model weights,

i) the factsheet PDF and the LaTeX source `.zip` attached.

Deadline. Tuesday 3 November 2026, 23:59 AoE (Anywhere on Earth), which is Wednesday 4 November 2026, 12:59 Berlin time. Factsheets received after the deadline will not be considered.

How the factsheet is used. The factsheet feeds the joint MaCVi 2027 challenge summary paper, where verified teams are listed and their authors become coauthors. A separate archival-paper window for individual solutions has not been confirmed.

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1 Team details

- **Team name (as on macvi.org):** [team name]
- **Team leader name:** [name]
- **Team leader institution and email:**
  [institution, city, country]
  Email: [leader e-mail]
- **Team leader's macvi.org user ID:** [ID]
- **Rest of the team members (name, macvi.org user ID):**
  - [member 1], user ID [ID]
  - [member 2], user ID [ID]
  - [. . .]
- **Affiliations:** [affiliations]
- **Author list for the summary paper, in your team's author order** (full name, affiliation, e-mail, OpenReview profile ID):
  1. [Full Name], [affiliation], [e-mail], [~Full_Name1]
  2. [Full Name], [affiliation], [e-mail], [~Full_Name1]
  3. [. . .]
- **Team website URL (if any):** [URL]
- **Final entry (the upload selected as final entry in the dashboard):** submission ID [ID], uploaded by user ID [ID], score shown on the leaderboard [mean relative-position drift, % of segment length]
- **Code repository and model weights:** [repository URL (public, or private and shared with the organizers); weights URL if not in the repository; access status and expiry date of each link]
- **Artifact traceability:** code commit [hash]; final artifact (the JSON bundle you uploaded) file name [name]; SHA-256 of the final artifact [hash]

2 Global method description

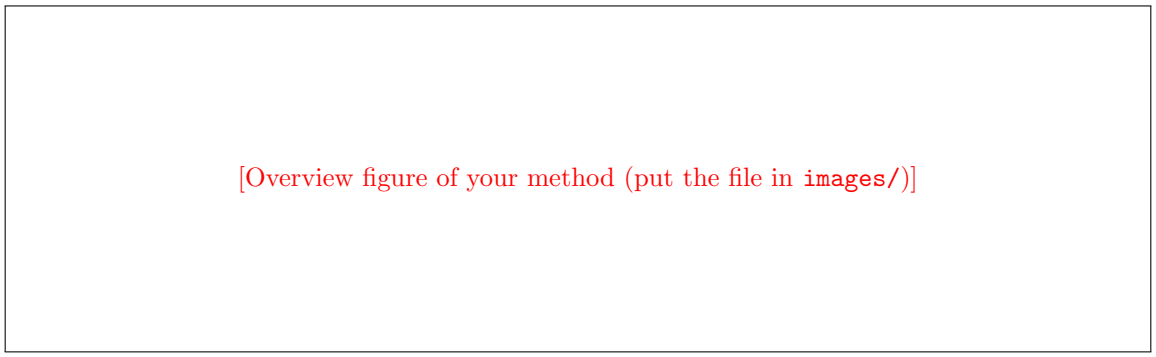


Figure 1: [Overview of the proposed method: the main blocks and the information flow from the monocular video frames to the timestamped camera trajectory.]

[General method description: the architecture or pipeline, how motion is estimated from the monocular video, how camera calibration and frame timestamps are used, what is new compared with published work. Add equations if useful. Give credit to all works your solution builds on, for example DPVO [1].]

- **Pretrained or external methods / models used (at any stage, if any):** [list them with citations and checkpoint names, or state "None"]
- **Additional data used in addition to the provided training split (at any stage, if any):** [dataset names, citations, size, how they were used; or "None"]
- **Training description:** [framework, optimizer, learning-rate schedule, batch size, number of iterations or epochs, input resolution, frame rate or stride, augmentations, losses and weights, hardware, training time; or "no training" if you used pretrained weights only]
- **Testing description:** [input resolution and frame rate at inference, scale recovery (if any), how tracking failures and lost or diverged segments are handled, any fallback used for clips where VO fails (state which clips and what the fallback uses), smoothing or post-processing, hardware]
- **How the development clips were used:** [which clips (IDs) were used for training, for validation and for model selection]
- **Compliance with the challenge rules:** [Yes / No; confirm that inference used only the monocular video, the camera calibration and the decoded frame timestamps (no GPS, IMU, AIS, speed, heading or other clip-specific navigation signals), and that no validation or test videos were used for training, including self-supervised training]
- **Were the released test frames used in any way for training, fine-tuning, adaptation or tuning of the method?** [Yes / No; if Yes, explain exactly how]
- **Which settings (seed, frame rate, resolution, thresholds, fallback rules, ...) were chosen using leaderboard scores?** [list each choice with its fixed value, applied identically to every clip; or "None"]
- **Novelty degree of the solution and whether it has been previously published:** [text]
- **Other methods and baselines tested (even if results were not competitive):** [text]

Method	Input size	Frame rate	Pretrained	Extra data	# Params. (M)	GPU
[name]	[H×W]	[fps]	[Yes/No]	[Yes/No]	[M]	[GPU]

Table 1: Summary of the method used for the final entry.

3 Implementation and environment

Give the training details once, in Section 2; do not repeat them here.

- **Programming language and framework (with versions):** [e.g. Python, PyTorch, CUDA]
- **Operating system and hardware used for training and for the reproduction run:** [OS, GPU/CPU, memory]
- **External code and dependencies (cite all external code):** [repositories with commits, licences, link to a dependency lock file such as `requirements.txt` or `environment.yml`]

4 Reproduction procedure

- **Installation commands:** [exact commands, in order]
- **Checkpoint file name(s) and SHA-256 hash(es):** [name, hash]
- **Inference command:** [the exact command(s) that produce the JSON bundle of your final entry from the released test frames, including packing the 20 clip predictions into one bundle, or point to the section of the README]
- **Expected output:** [output path and file name of the JSON bundle; expected SHA-256 if your pipeline is deterministic]
- **Fixed seeds and any nondeterministic operations:** [seeds, nondeterministic GPU operations, and whether repeated runs give byte-identical output]
- **Expected reproduced score:** [the mean relative-position drift (% of segment length) you expect from the reproduction, and the run-to-run variation you observed]
- **Run time and hardware of the full reproduction:** [minutes for all 20 test clips, GPU/CPU]

5 Other details

- **Planned submission of a solution description paper to the MaCVi@WACV 2027 workshop:** [Yes / No]
- **General comments and impressions of the challenge (we appreciate your feedback to improve future editions):** [text]
- **Known limitations and failure cases of the method:** [text, e.g. low-texture open water, low parallax, thermal footage]

Declaration

I confirm that the reported final entry was produced using only the monocular video, the camera calibration and the decoded frame timestamps as inference inputs; that all external datasets, pretrained models, ensembles, fallbacks and post-processing are disclosed in this factsheet; that the submitted code and weights correspond to the named final entry; and that the method complies with the official Maritime Visual Odometry rules.

Team leader: [name] Date: [date]

References

[1] Zachary Teed, Lahav Lipson, and Jia Deng. Deep patch visual odometry. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.